Band-1 Hadamard Order 3076 — Miyamoto (H_768 Paley-I×2); Miyamoto-Only Bonus; Gap 2052 Still OPEN

H_384/H_768/C_770/H_3076 PASS (Miyamoto); 3076 NOT in SPK (Paley-I/II/Kronecker FAIL); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_3044_Miyamoto`; `2026-09-15_Band1_Hadamard_2836_Miyamoto`; `2026-09-15_Band1_Hadamard_2804_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_3076_miyamoto/`.
Cutoff. 2026-09-15 (~04:42:20Z; ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 384, 768, and 3076 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-I seed H_{384} Sylvester-doubled to H_{768}. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 3076 is a **Miyamoto-only bonus** (**not** in SPK): honest SPK audit — Paley-I **FAIL** ($q = 3075$ not prime); Paley-II **FAIL** ($q = 1537$ not prime); Kronecker **FAIL** ($3076 = 4 \cdot 769$, 769 not Hadamard-admissible). Paley-II path for seed H_{768} **REJECTED** ($768 = 2(q+1) \Rightarrow q = 383 \equiv 3 \pmod 4$). Kronecker($4 \times 192$) alt for H_{768} noted but unused (Paley-I double preferred). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 3044 on box. This increment constructs the **next preferred** Miyamoto-eligible target after $q = 761$: **order 3076** ($3076 = 4 \cdot 769$, 769 prime $\equiv 1 \pmod 4$). Seed H_{384} via Paley type I ($q = 383$ prime $\equiv 3 \pmod 4$; order $q + 1 = 384$), then Sylvester double H_{768} (Paley-II path for 768 **rejected**; Kronecker 4×192 alt noted). Symmetric Paley conference C_{770} over prime 769. $H_{3076} = \text{Miyamoto}(H_{768}, C_{770})$. Unit tests **29/29 PASS**. Order 3076 does **not** enter SPK (Miyamoto-only bonus). SPK next gap remains **2052** (OPEN). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 3076; Miyamoto; Paley type I; Sylvester doubling; Paley conference; Miyamoto-only bonus; constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{384} constructed and verified (Paley type I, $q = 383$). 2. H_{768} constructed and verified (Sylvester double of H_{384}). Paley-II path **rejected**. Kronecker(4×192) alt noted. 3. C_{770} constructed and verified (Paley conference, $q = 769$): $CC^\top = 769I$, zero diagonal. 4. H_{3076} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. SPK: $3076 \notin$ SPK (Miyamoto-only bonus). Paley-I, Paley-II, and Kronecker paths for 3076 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$3076 = 4 \cdot 769$
$q = 769$	prime, $769 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{768}
Needed conference	order $770 = q + 1$
H_{384} path	Paley type I, $q = 383$ (prime $\equiv 3 \pmod 4$); $q + 1 = 384$
H_{768} path (used)	Sylvester double of H_{384}
Paley-II for H_{768}	**REJECTED**: $768 = 2(q + 1) \Rightarrow q = 383 \equiv 3 \pmod 4$ (needs $\equiv 1$)
Kronecker alt for H_{768}	Noted: 4×192 (unused; Paley-I double preferred)
Conference path	Symmetric Paley over prime 769
Paley-I for 3076	**REJECTED**: $q' = 3075$ not prime
Paley-II for 3076	**REJECTED**: $q' = 1537$ not prime

Quantity	Value
Kronecker for 3076	**REJECTED**: only 4×769; $769 \not\equiv 0 \pmod 4$ (not admissible)
SPK membership	**False** (Miyamoto-only bonus)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{384}	Paley type I, $q = 383$	0	0.1023	**PASS**
H_{768}	Sylvester double of H_{384}	0	0.2111	**PASS**
C_{770}	Paley conference, $q = 769$	0	0.4603	**PASS**
H_{3076}	Miyamoto(H_{768}, C_{770})	0	12.7132	**PASS**

Primary Miyamoto pipeline wall (incl. unit tests + SPK refresh) $\approx$ 14.0319 s. Compressed ± 1 matrices saved as `H384_pm1.npz`, `H768_pm1.npz`, `C770_pm1.npz`, `H3076_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior ships (incl. 380, 760, 3044, 708, 2836, 700, 2804) $\cup$ verified H_{384}, $H_{768} \cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 3076 therefore remains **outside** SPK.

Quantity	Value
Known count	884
3076 in SPK	**False** (Miyamoto-only bonus)
768 in SPK	True (Sylvester double of Paley-I H_{384})
384 in SPK	True (Paley I $q = 383$)
3044 in SPK	False (prior Miyamoto bonus)
2836 in SPK	False (prior Miyamoto bonus)
2804 in SPK	True (prior Paley I)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, 2148, 2156, 2172, 2212, 2284…

5. Gap 2052 (still OPEN; non-finding ONLY)

This ship does **not** fill gap 2052. Prior triage stands: Paley-I/II / Kronecker / Miyamoto-Paley-conf all **FAIL** for order 2052. Claim level for 2052 remains method triage / non-finding ONLY. Non-finding $\neq$ counterexample; Hadamard conjecture remains OPEN. No H_{2052} matrix is claimed.

6. Unit tests (29/29)

Test	Result
H4_sylvester	PASS
H20_paley_I_control	PASS
paley_II_q5_H12	PASS
paley_II_q13_H28	PASS
miyamoto_H20_from_H4_conf6	PASS
paley_I_q7_H8_double_H16	PASS
paley_I_q11_H12	PASS
q383_is_prime	PASS
q383_equiv_3_mod_4	PASS
H384_order_formula	PASS
H768_double_formula	PASS
q769_is_prime	PASS
q769_equiv_1_mod_4	PASS
n3076_equals_4_times_769	PASS
H768_order_formula	PASS
C770_order_formula	PASS
paley_II_H768_path_rejected	PASS
paley_I_H3076_path_rejected	PASS
paley_II_H3076_path_rejected	PASS
kronecker_H3076_path_rejected	PASS
q3075_not_prime	PASS
q1537_not_prime	PASS
reject_all_ones_H8	PASS
conference_6_paley_control	PASS
miyamoto_eligible_3076_arith	PASS
conference_770_shape_expected	PASS
gap_2052_not_filled_by_this_ship	PASS
3076_miyamoto_only_bonus_arith	PASS
H768_kronecker_4x192_shape_note	PASS

7. Next optional targets

Miyamoto-eligible primes $q > 769$, $q \equiv 1 \pmod 4$ (sample):

q	$n = 4q$	Need H_{q-1}	On-box path notes
773	3092	772	—
797	3188	796	PaleyII(q=397)

q	$n = 4q$	Need H_{q-1}	On-box path notes
809	3236	808	Kronecker(404x2)
821	3284	820	PaleyII(q=409)
829	3316	828	PaleyI(q=827)
853	3412	852	—
857	3428	856	Kronecker(428x2)
877	3508	876	—

Preferred next. Method-attack remaining SPK gap **2052** (GS/SDS for order ~513 if literature seeds appear), **or** construct Miyamoto $q = 797 \to n = 3188$ (Paley-II $q = 397$), **or** $q = 829 \to n = 3316$ (Paley-I $q = 827$), **or** Track B privacy / Object 1 meter await. GATE-04 remains LOCKED.

8. Non-claims (explicit)

- Not the full Hadamard conjecture.
- Not Clay-closed / finished-ToE / RH / P≠NP / YM / Frankl / NS-solved / AC-solved.
- Gap **2052** remains **OPEN**.
- Order 3076 is **not** in SPK (Miyamoto-only bonus).
- GATE-04 collection **LOCKED**; no invented J.
- `clay_closed = false`.

9. Reproducibility

```
cd /workspace/experiments/band1/hadamard_3076_miyamoto
python3 verify_hadamard_3076.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `H384_pm1.npz`, `H768_pm1.npz`, `C770_pm1.npz`, `H3076_pm1.npz`.